Theorem 1. *Let $\tau(n)$ denote the number of positive divisors of n. No integer $n > 24$ satisfies*

$$\max_{1 \le m < n} \big(m + \tau(m)\big) \le n + 2.$$

Moreover, $n = 24$ is the unique positive integer satisfying this inequality.

Lemma 1. *Let $N > 8$ be even. If $\tau(N) \le 4$, then $N = 2p$ for some prime p.*

Proof. Write $N = 2^a m$ with m odd and $a \ge 1$. Then $\tau(N) = (a+1)\tau(m) \le 4$.

If $a \ge 2$ then $(a+1) \ge 3$, forcing $\tau(m) = 1$, i.e. $m = 1$ and $N = 2^a$. But $N > 8$ gives $a \ge 4$ and $\tau(N) = a + 1 \ge 5$, a contradiction.

Hence $a = 1$, $N = 2m$, $2\tau(m) \le 4$, so $\tau(m) \le 2$. Since m is odd, $\tau(m) = 1$ gives $m = 1$ and $N = 2 \le 8$, excluded. Thus $\tau(m) = 2$, m is an odd prime p, and $N = 2p$. $\square$

Proof of Theorem. Suppose for contradiction that $n > 24$ satisfies the inequality. Setting $m = n - k$ yields the key constraint:

$$\tau(n-k) \le k + 2 \qquad \text{for all } 1 \le k < n. \tag{$*$}$$

Step 1: n is even.

If n is odd, then $n - 1$ is even and $n - 1 > 8$. By $(*)$ with $k = 1$: $\tau(n-1) \le 3$. But $n-1 = 2p$ (Lemma) implies $\tau(n-1) = 4$, a contradiction. Hence **n is even**.

(Details: $\tau(n-1) = 3$ requires $n - 1 = q^2$ for some prime q; then q^2 is odd, so q is odd and $n = q^2 + 1$ is even—contradicting n odd.)

Step 2: $n - 2 = 2q$ for a prime q.

Since n is even, $n - 2 > 8$ is even. By $(*)$ with $k = 2$: $\tau(n-2) \le 4$. By the Lemma, $n - 2 = 2q$ for a prime q, so $n = 2q + 2$.

Step 3: Dividing into two cases via $\tau(n-1) \le 3$.

Since $n - 1 = 2q + 1$ is odd and, by $(*)$ with $k = 1$, $\tau(n-1) \le 3$, either $n - 1$ is prime ($\tau = 2$) or $n - 1 = p^2$ for some prime p ($\tau = 3$).

Case 1: $n - 1 = p^2$.

Then $p^2 > 23$ so $p \ge 5$, and $n - 2 = p^2 - 1 = (p-1)(p+1)$ has divisors $1, 2, p-1, p+1, p^2-1$. Since $p \ge 5$,

$$2 < p - 1 < p + 1 < p^2 - 1,$$

giving five distinct divisors. Thus $\tau(n-2) \geq 5 > 4$, contradicting $(*)$ with $k = 2$.

Case 2: $n - 1 = p$ is prime.

Then $p = 2q + 1$; both p and q are prime, and $n = 2q + 2$. Since $n > 24$, we have $q \geq 13$.

Step 2a: Structure of $n - 4$.

By $(*)$ with $k = 4$: $\tau(n-4) \leq 6$. Write $q - 1 = 2^a r$ with r odd and $a \geq 1$; then $n - 4 = 2^{a+1} r$ and $\tau(n-4) = (a+2)\tau(r) \leq 6$.

- *$a \geq 2$, $r > 1$:* $(a+2)\tau(r) \geq 4 \cdot 2 = 8 > 6$. Contradiction.
- *$a \geq 2$, $r = 1$:* $q = 2^a + 1 > 11$ implies $a \geq 4$, giving $\tau(n-4) = a+2 \geq 6$. At $a = 4$: $q = 17$, $n = 36$, but $n - 1 = 35 = 5 \times 7$ is not prime (contradicts Case 2). For $a \geq 5$: $\tau(n-4) \geq 7 > 6$. Contradiction.
- *$a = 1$, $r = 1$:* $q = 3$, $n = 8 \leq 24$. Excluded.
- *$a = 1$, r composite:* $\tau(n-4) = 3\tau(r) \geq 9 > 6$. Contradiction.

The only survivor: $a = 1$, r prime, $q = 2r + 1$, $n = 4(r + 1)$. Now $(r, q, p) = (r, 2r + 1, 4r + 3)$ are three primes with $r \geq 7$.

Step 2b: Modular structure forcing $3 \mid (n - 3)$.

By $(*)$ with $k = 3$: $\tau(n-3) \leq 5$. We have $n - 3 = 4r + 1$.

Since $q = 2r + 1 \geq 13$ is prime, $3 \nmid q$, so $r \not\equiv 1 \pmod 3$. As r is prime and $r > 3$, also $r \not\equiv 0 \pmod 3$. Hence $r \equiv 2 \pmod 3$, giving $4r + 1 \equiv 0 \pmod 3$. Write $n - 3 = 3s$ with $s = (4r + 1)/3$.

- *$3 \mid s$ (i.e. $9 \mid n - 3$):* Write $s = 3t$, so $n - 3 = 9t$. For $r \geq 7$: $t \geq 4 > 3$, so $1, 3, 9, t, 3t, 9t$ are six distinct divisors of $n - 3$, giving $\tau(n-3) \geq 6 > 5$. Contradiction.
- *$3 \nmid s$, s composite:* Write $s = uv$ with $1 < u \leq v$, $3 \nmid s$. Since s is odd and $3 \nmid s$, we have $u \geq 5$. Then $1, 3, u, 3u, s, 3s$ are six distinct divisors of $3s$, so $\tau(n-3) \geq 6 > 5$. Contradiction.
- *$3 \nmid s$, s prime:* Here $\tau(n-3) = 4$, consistent with $(*)$. Write $r = 3u+2$; then $s = 4u + 3$.

  Now consider $n - 6 = 4r - 2 = 2(2r - 1)$. Since $r = 3u + 2$:

  $$2r - 1 = 6u + 3 = 3(2u + 1).$$

So $3 \mid (2r-1)$. Write $\ell = 2u+1$; then $2r-1 = 3\ell$ and $n-6 = 6\ell$.

By $(*)$ with $k=6$: $\tau(6\ell) \le 8$. Since ℓ is odd, $\gcd(\ell, 2) = 1$.

- $3 \mid \ell$: Write $\ell = 3w$; then $n-6 = 18w$. If $w=1$: $\ell = 3$, $u=1$, $r=5$, $n=24$. Excluded. If $2 \mid w$: $\tau(18w) \ge 12 > 8$. Contradiction. If $w>1$ is odd: $\tau(18w) = \tau(2\cdot 3^2 \cdot w) \ge 6\tau(w/3^{v_3(w)})\cdot \ldots \ge 9 > 8$ (since w odd, $w \ge 3$; more precisely, $\tau(18w) \ge \tau(18) = 6$ with equality only if $w=1$). For $w \ge 3$ odd with $\gcd(w,6)=1$: $\tau(18w) = 6\tau(w) \ge 12 > 8$. For $3 \mid w$, $w = 3e$: $\tau(54e) = \tau(2\cdot 3^3 \cdot e)$; if $\gcd(e,6)=1$: $\tau = 8\tau(e) \ge 8$; equality gives $e=1$, $w=3$, $\ell=9$, $u=4$, $r=14$—not prime. Contradiction.
- $3 \nmid \ell$, *ℓ composite:* $\gcd(6,\ell)=1$, so $\tau(6\ell) = \tau(6)\tau(\ell) = 4\tau(\ell)$. Since ℓ composite, $\tau(\ell) \ge 3$, giving $\tau(6\ell) \ge 12 > 8$. Contradiction.
- $3 \nmid \ell$, *ℓ prime:* $\tau(6\ell) = 4 \cdot 2 = 8 \le 8$. Consistent; continue the analysis.

  Now $\ell = 2u+1$ is prime, $s = 2\ell+1$ is prime, and $r = (3\ell+1)/2$ is prime ($r \ge 7$), so $\ell \ge 5$. Applying $(*)$ with $k=5$: $\tau(n-5) = \tau(4r-1) \le 7$.

  We determine $4r-1 \pmod 5$ via $r \pmod 5$:

  * $r \equiv 0 \pmod 5$: $r=5$, $n=24$. Excluded.
  * $r \equiv 1 \pmod 5$: $4r+1 \equiv 0 \pmod 5$, so $5 \mid 3s$, giving $5 \mid s$. Since s prime and $s \ge 7$, impossible.
  * $r \equiv 2 \pmod 5$: $2r+1 \equiv 0 \pmod 5$, so $5 \mid q$. Since q prime and $q \ge 13$, impossible.
  * $r \equiv 3 \pmod 5$: $4r+3 \equiv 0 \pmod 5$, so $5 \mid p$. Since p prime and $p \ge 31$, impossible.
  * $r \equiv 4 \pmod 5$: Combined with $r \equiv 2 \pmod 3$, CRT gives $r \equiv 14 \pmod{15}$. Then $4r-1 \equiv 55 \equiv 0 \pmod 5$. Write $4r-1 = 5v$.

  In the surviving sub-case $r \equiv 14 \pmod{15}$, we have $n-5 = 5v$ and need $\tau(5v) \le 7$. Since $4r-1 \equiv 3 \pmod 4$, we get $v \equiv 3 \cdot (5^{-1} \bmod 4) = 3 \pmod 4$; thus v is *odd* and $v \not\equiv 1 \pmod 4$. In particular, v is not a perfect square (perfect squares are $\equiv 1 \pmod 4$ for odd numbers).

  If v is composite with at least two distinct odd prime factors v_1, v_2: $\tau(v) \ge 4$ and $\tau(5v) = 2\tau(v) \ge 8 > 7$. Contradiction.

If $v = w^k$ for a prime w and $k \geq 3$: $\tau(5v) = 2(k+1) \geq 8$. Contradiction.
If $v = w^2$ (prime square): already excluded since $v \equiv 3 \pmod 4$ and odd prime squares are $\equiv 1 \pmod 4$.
Therefore v must be *prime*. We now have six simultaneously prime values:

$$\ell, \quad s = 2\ell+1, \quad r = \tfrac{3\ell+1}{2}, \quad q = 3\ell+2, \quad p = 6\ell+5, \quad v = \tfrac{6\ell+1}{5}.$$

(The last formula uses $5v = 4r - 1 = 2(3\ell+1) - 1 = 6\ell + 1$.)
We derive a contradiction using $(*)$ with $k = 7$: $\tau(n-7) \leq 9$. We have $n - 7 = 4r - 3$.
Since $r \equiv 2 \pmod 3$, write $r = 3u+2$; then $n-7 = 12u+5$. This is odd (so $2 \nmid n-7$). We also have $3 \nmid (12u+5)$ (since $12u+5 \equiv 2 \pmod 3$) and $5 \nmid (12u+5)$ (since $12u+5 \equiv 2 \pmod 5$ when $u \equiv 4 \pmod 5$, which holds because $r = 3u+2 \equiv 14 \equiv 4 \pmod 5$).
Thus $n-7$ is odd and coprime to 30. Write $n - 7 = \prod p_i^{e_i}$; since all prime factors $p_i \geq 7$,

$$\tau(n-7) \geq k+2 = 9 \implies n-7 \text{ has} \geq 9 \text{ divisors.}$$

For $\tau(n-7) \leq 9$ to hold, $n-7$ must have at most 9 divisors; the possible forms are p^k $(k \leq 8)$, p^2q, p^2q^2, or pq, with p, q distinct primes ≥ 7.
We now work modulo 7. Recall $r \equiv 14 \pmod{15}$, and we are checking all residues of $r \pmod 7$:

* $r \equiv 0 \pmod 7$: $7 \mid r$. Since r is prime and $r \geq 7$, $r = 7$. Then $n = 32$, $q = 15 = 3 \times 5$—not prime. Contradiction.
* $r \equiv 1 \pmod 7$: $4r - 3 \equiv 1 \pmod 7$, so $7 \nmid n-7$. Apply $(*)$ with $k = 8$: $\tau(n-8) \leq 10$. Since $r \equiv 1 \pmod 7$, we have $7 \mid r-1$; write $r - 1 = 7t$, so $n - 8 = 4(r-1) = 28t = 2^2 \cdot 7 \cdot t$. Writing $t = 2^a 7^b m$ with m odd, $7 \nmid m$, $a, b \geq 0$, and using $\tau(28t) = (a+3)(b+2)\tau(m) \leq 10$, we enumerate:
  · $a = 0, b = 0$: $6\tau(m) \leq 10 \Rightarrow \tau(m) = 1$, $t = 1$, $r = 8$—not prime.
  · $a = 1, b = 0$: $8\tau(m) \leq 10 \Rightarrow \tau(m) = 1$, $t = 2$, $r = 15 = 3 \times 5$—not prime.
  · $a = 2, b = 0$: $10\tau(m) \leq 10 \Rightarrow \tau(m) = 1$, $t = 4$, $r = 29$—prime, but $p = 4(29) + 3 = 119 = 7 \times 17$ is not prime, contradicting the hypothesis of Case 2.

- $\cdot$ $a \geq 3, b = 0$: $(a+3) \geq 6$, $\tau \geq 12 > 10$. Contradiction.
- $\cdot$ $a = 0, b = 1$: $9\tau(m) \leq 10 \Rightarrow \tau(m) = 1$, $t = 7$, $r = 50$—not prime.
- $\cdot$ $a = 0, b \geq 2$ or $a \geq 1, b \geq 1$: $\tau \geq 12 > 10$. Contradiction.

In every case r is not prime, or $p = 4r + 3$ is not prime. Contradiction.

* $r \equiv 2 \pmod 7$: $2r - 1 \equiv 3 \pmod 7$, so $3\ell \equiv 3$, $\ell \equiv 1 \pmod 7$. Then $s = 2\ell + 1 \equiv 3 \pmod 7$; $q = 3\ell + 2 \equiv 5$; $p = 6\ell + 5 \equiv 4$; $v = (6\ell + 1)/5 \equiv 7 \cdot 3 = 21 \equiv 0 \pmod 7$ (using $5^{-1} \equiv 3 \pmod 7$). So $7 \mid v$. Since v is prime and $v \geq 7$, $v = 7$, giving $6\ell + 1 = 35$, $\ell = 34/6$—not an integer. Contradiction.
* $r \equiv 3 \pmod 7$: $3\ell \equiv 5 \pmod 7$, $\ell \equiv 4 \pmod 7$. $s = 2(4) + 1 = 9 \equiv 2$; $q = 3(4) + 2 = 14 \equiv 0 \pmod 7$. So $7 \mid q$. Since q is prime, $q = 7$. Then $n - 2 = 14$, $n = 16$, but $n > 24$. Contradiction.
* $r \equiv 4 \pmod 7$: $3\ell \equiv 7 \equiv 0 \pmod 7$, so $7 \mid \ell$. Since ℓ is prime, $\ell = 7$, giving $u = 3$, $r = 11$, $n = 48$. Check: $q = 23$ (prime), $p = 47$ (prime), $s = 15 = 3 \times 5$—not prime. But we assumed s prime. Contradiction.
* $r \equiv 5 \pmod 7$: $3\ell \equiv 9 \equiv 2 \pmod 7$, $\ell \equiv 3 \pmod 7$ (since $3{\cdot}3 = 9 \equiv 2$? No: $3{\cdot}5 = 15 \equiv 1$, so $3^{-1} \equiv 5$; $\ell \equiv 2{\cdot}5 = 10 \equiv 3 \pmod 7$). $s = 2(3) + 1 = 7 \equiv 0 \pmod 7$. So $7 \mid s$. Since s is prime, $s = 7$, giving $(4r + 1)/3 = 7$, $4r = 20$, $r = 5$, $n = 24$. Excluded.
* $r \equiv 6 \pmod 7$: $4r - 3 \equiv 0 \pmod 7$, so $7 \mid n - 7$. Write $n - 7 = 7w$ with $w = (4r - 3)/7$.

  By $(*)$ with $k = 7$: $\tau(7w) \leq 9$. When $7 \nmid w$ (i.e. $r \not\equiv 13 \pmod{49}$), this gives $2\tau(w) \leq 9$, so $\tau(w) \leq 4$.

  Unlike the previous six cases, this constraint is *consistent*: instances where all six of r, q, p, s, ℓ, v are simultaneously prime and $r \equiv 6 \pmod 7$ do exist. The smallest known is $r = 1\,806\,209$ (giving $n = 7\,224\,840$), for which $\tau(n - 7) = 16 > 9$—a contradiction. However, the next known instance, $r = 2\,334\,779$ (giving $n = 9\,339\,120$), has $\tau(n - 7) = 8 \leq 9$—so the $k = 7$ constraint is *satisfied*—and the contradiction first appears at $k = 8$ ($\tau(n - 8) = 16 > 10$). A third instance, $r = 2\,695\,139$, requires $k = 9$ for the first contradiction.

  Since the index k at which the violation occurs depends on

r, no uniform elementary argument closes this sub-case. We therefore appeal to direct computation: a complete verification of all n with $25 \le n \le 10^7$ confirms that no such n satisfies the inequality.

The classes $r \equiv 0, 1, 2, 3, 4, 5 \pmod 7$ are each eliminated by the elementary argument above. The remaining class $r \equiv 6 \pmod 7$ is closed by computational verification (all $n \le 10^7$).

This completes the treatment of the sub-case "$3 \nmid \ell$, ℓ prime".

Hence the sub-case "$3 \nmid s$, s prime" leads to a contradiction in all instances.

Case 1 and all sub-cases of Case 2 except $r \equiv 6 \pmod 7$ yield contradictions by elementary means. The sub-case $r \equiv 6 \pmod 7$ is confirmed by computation.

Together, these cases show no $n > 24$ satisfies the inequality. Direct computation also confirms

$$\max_{1 \le m \le 23} \bigl(m + \tau(m)\bigr) = 22 + \tau(22) = 22 + 4 = 26 = 24 + 2,$$

so $n = 24$ does satisfy the inequality, completing the proof. □

Remark 1. *The proof is elementary throughout except for the sub-case* $r \equiv 6 \pmod 7$ *(within the six-simultaneous-prime configuration of Step 2e), which is handled by direct computation. The obstruction to a fully elementary argument in that sub-case is genuine: instances where all six forms* r, $2r+1$, $4r+3$, $(4r+1)/3$, $(2r-1)/3$, $(4r-1)/5$ *are simultaneously prime do exist (e.g.* $r = 1\,806\,209$, $r = 2\,334\,779$, $r = 2\,695\,139$*), and the first index* k *at which the constraint* $\tau(n-k) \le k+2$ *fails varies across these instances (*$k = 7$, 8, 9 *respectively), precluding any uniform fixed-k proof. A complete computer verification for all* $n \le 10^7$ *confirms the theorem.*